

CODE

```
// Include NewPing Library

#include "NewPing.h"

// Hook up HC-SR04 with Trig to Arduino Pin 9, Echo to Arduino pin 10

#define TRIGGER_PIN 9

#define ECHO_PIN 10

// Maximum distance we want to ping for (in centimeters).

#define MAX_DISTANCE 400

// NewPing setup of pins and maximum distance.

NewPing sonar(TRIGGER_PIN, ECHO_PIN, MAX_DISTANCE);

float duration, distance;

void setup()

{

  Serial.begin(9600);

}

void loop()

{

  // Send ping, get distance in cm

  distance = sonar.ping_cm();
```

```
// Send results to Serial Monitor

Serial.print("Distance = ");

if (distance >= 400 || distance <= 2)
{
    Serial.println("Out of range");
}
else
{
    Serial.print(distance);
    Serial.println(" cm");
}

delay(500);
}

// includes the LiquidCrystal Library
#include <LiquidCrystal.h>

// includes the LcdBarGraph Library
#include <LcdBarGraph.h>

// Maximum distance we want to ping for (in centimeters).
#define max_distance 200
```

```
// Creates an LCD object. Parameters: (rs, enable, d4, d5, d6, d7)
```

```
LiquidCrystal lcd(12, 11, 5, 4, 3, 2);
```

```
LcdBarGraph lbg(&lcd, 16, 0, 1); // Creates an LCD Bargraph object.
```

```
const int trigPin = 9;
```

```
const int echoPin = 10;
```

```
long duration;
```

```
int distance;
```

```
void setup()
```

```
{
```

```
  lcd.begin(16,2); // Initializes the interface to the LCD screen
```

```
  pinMode(trigPin, OUTPUT);
```

```
  pinMode(echoPin, INPUT);
```

```
}
```

```
void loop()
```

```
{
```

```
  // Write a pulse to the HC-SR04 Trigger Pin
```

```
  digitalWrite(trigPin, LOW);
```

```
  delayMicroseconds(2);
```

```
  digitalWrite(trigPin, HIGH);
```

```
delayMicroseconds(10);

digitalWrite(trigPin, LOW);


// Measure the response from the HC-SR04 Echo Pin

duration = pulseIn(echoPin, HIGH);


// Determine distance from duration

// Use 343 metres per second as speed of sound

distance= duration*0.034/2;


// Prints "Distance: <value>" on the first line of the LCD

lcd.setCursor(0,0);

lcd.print("Distance: ");

lcd.print(distance);

lcd.print(" cm");


// Draws bargraph on the second line of the LCD

lcd.setCursor(0,1);

lbg.drawValue(distance, max_distance);

delay(500);

}
```